

Sticky Information versus Sticky Prices

A proposal to replace the New Keynesian Philips Curve

based on Mankiw & Reis (2002)

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The classical New Keynesian Phillips Curve

- The paper was written not long after introduction of New Keynesian Philips Curve.
- The model is not yet fully developed - it is before Woodford (2003) or Galí (2008).
- However, a few important elements have already emerged, and generally agreed on the existence of price stickiness and resulting Taylor curve

Possible problems

- 1 Lack of "long and variable shocks"
- 2 Not really working with data - lacking "inflation inertia"
- 3 Disinflationary booms
- 4 Does not survive McCallum critique

Price stickiness

The authors choose the Calvo style price stickiness: each period, a fraction λ of the firms with perfect information are given an opportunity to adjust their prices. The desired price is given by the equation:

$$p_t^* = p_t + \alpha y_t$$

The adjustment price is therefore given by: x

$$x_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_t p_{t+j}^*$$

and the aggregate price level:

$$p_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j x_{t-j}$$

Information stickiness

The authors' proposal is based on different idea: firms are free to select their prices, but they are working (rationally) on the past information, and only with Calvo probability λ do they get new information. The wanted price is:

$$p_t^* = p_t + \alpha y_t$$

but a firm who selected price j periods ago has chosen:

$$x_t^j = \mathbf{E}_{t-j} p_t^*$$

The aggregate price level is still:

$$p_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j \mathbf{E}_{t-j} x_t^j$$

Phillips curves

By simple arithmetic, the price-sticky Phillips Curve is as follows:

$$\pi_t = \frac{\alpha\lambda^2}{1-\lambda}y_t + \mathbf{E}_t\pi_{t+1}$$

For the sticky information Phillips Curve:

$$\pi_t = \frac{\alpha\lambda^2}{1-\lambda}y_t + \lambda \sum_{j=0}^{\infty} (1-\lambda)^j \mathbf{E}_{t-j-1}(\pi_t + \alpha g_t)$$

As a benchmark, the alternative expectations: backward-looking, have been adopted:

$$\pi_t = \frac{\alpha\lambda^2}{1-\lambda}y_t + \pi_{t-1}$$

Dynamics of the economy

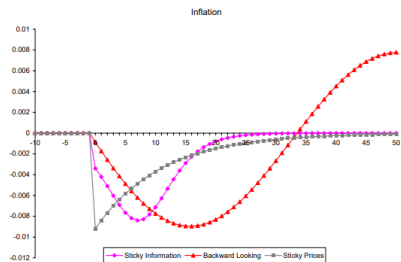
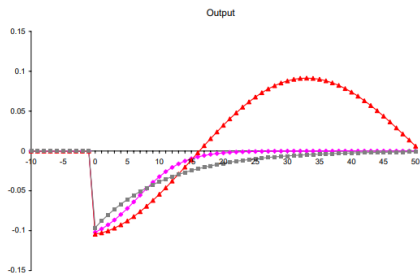
The demand function is assumed to be in form of:

$$m_t = y_t + p_t$$

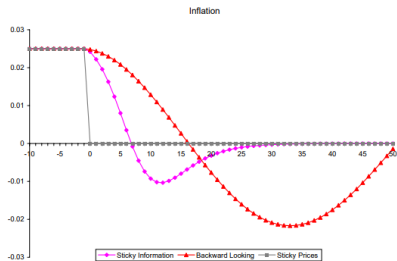
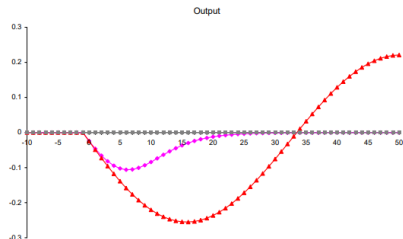
with possible interpretation of either QTM or some demand function. m_t is assumed to be exogenous process, with which monetary policy may interact. The desired price is therefore in a form:

$$p_t^* = (1 - \alpha)p_t + \alpha m_t$$

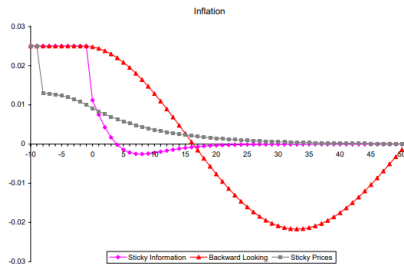
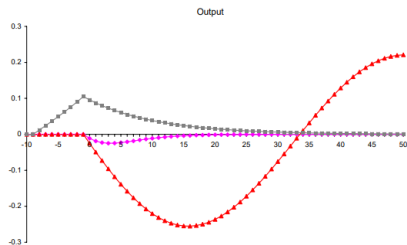
Reaction to fall in demand (10% fall in m_t)



Reaction to sudden disinflation (sudden stop of growth of m_t)



Reaction to announced disinflation (announced stop of growth of m_t)



Comparison with data

	$\text{corr}(y_t, \pi_{t+2} - \pi_{t-2})$	$\text{corr}(y_t, \pi_{t+4} - \pi_{t-4})$
<u>A. Actual</u>		
GDP deflator	.48	.60
CPI	.38	.46
core CPI	.46	.51
<u>B. Predicted</u>		
Backward-Looking Model	.99	.99
Sticky-Price Model	-.13	-.11
Sticky-Information Model	.43	.40

Conclusion

- The sticky information is better suited to explain the inflation inertia
- Sticky prices alone work too fast in rational expectations
- Although a proposal was radical, the general effects of both were not that different
- Such a model is better than the previous approach to sticky information based on Fischer (1977)